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Al,F-doped new perovskite lithium fast ion conductor $\text{Li}_{3x}\text{La}_{2/3-x}\text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$)

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Abstract Al, F-doped new perovskite lithium ion conductors $\text{Li}_{3x}\text{La}_{2/3-x}\text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$) have been prepared by solid state reaction. It is found that a pure perovskite-structured phase with space group of P4mm(99) exists in the composition range of $0 < y \leq 0.10$. The sample with $y=0.02$ possesses the highest ionic conductivity of 1.06×10^{-3} S/cm at room temperature, and its decomposing voltage is 2.3 V. The factors affecting the conductivity of this system are discussed.

Keywords Perovskite · Lithium fast ion conductor · LLT · Ionic conductivity · Al, F-doped

Introduction

Lithium fast ion conductors are of great interest for their potential use as electrolyte of full solid-state batteries and in other electrochemical devices. Of these materials, $\text{Li}_{3x}\text{La}_{2/3-x}\text{TiO}_3$ (LLT), where $\square$ represents vacancy with a perovskite-related structure, has attracted peculiar attention for its high lithium ion conductivity of 1.0×10^{-3} S/cm at room temperature when $x \approx 0.10$ [1]. To further improve lithium ion conductivity, much effort has been dedicated to substitute A-site La by other cations, such as other rare earth cations or alkaline earth cations, and/or substitute B-site Ti by other trivalent, tetravalent, pentavalent or hexavalent cations [2–13]. So far, only doping Sr up to 5% [12] and partial substitution of Al for Ti [10] are reported to increase the ionic conductivity slightly.

Based on these research, the ionic conductivity is proposed to depend mainly on the following factors [1, 10–13]:

1. The size of the A-site cation. As lithium ion conductivity originates from the motion of lithium along the A-site vacancies, the window of four oxygens separating adjacent A sites constitute the bottleneck for lithium ion migration. Thus, La substituted by smaller ions reduces the bottleneck size and suppresses the lithium ion migration, while the substitution of larger ions increases the bottleneck size and enhances the ion migration.
2. Lithium ion and vacancy concentration in the A sublattice.
3. Ordering or disordering of La, Li, and vacancies. The more the arrangement of their distribution is ordered, the lower the conductivity will be.
4. The facility of tilting of TiO_6 octahedra. The easier the tilting the higher the ionic conductivity.
5. The local structure. The local lattice distortion will increase the activation energy of ionic conductivity and will decrease the ionic conductivity.
6. The change of bond strength between A–O and B–O bond in ABO_3 perovskites. Stronger B–O bond favors high ionic conductivity.

In this study, a new perovskite solid solution with F and Al co-doped LLT were synthesized for the first time and characterized by AC impedance spectroscopy, X-ray diffraction (XRD), and DC polarization.

Experimental description

Different compositions of the $\text{Li}_{3x}\text{La}_{2/3-x}\text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$) system were prepared from stoichiometric amounts of Li_2CO_3 , La_2O_3 , TiO_2 , Al_2O_3 , and LiF plus 5% extra weight of Li_2CO_3 to compensate lithium loss at high temperature. All the reagents were dried just before use. The mixture was ground together in agate mortar until

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thoroughly blended and then transferred to Pt boats and heated at 700 °C for 4 h for decarbonation. Afterwards, the samples were reground, pelletized, covered with powder of the sample of the same composition, and fired at 1,200 °C for 25 h to finish the reaction.

The crystal structure of the product was investigated by powder X-ray diffraction in Philips X'pert Pro diffractometer using Cu K α 1 ($\lambda=1.54056$ Å) radiation at room temperature.

Electrical conductivity was measured by an AC impedance technique on reheated discs (13-mm diameter and about 2-mm thick) from room temperature to 400 °C in the frequency range of 0.1 Hz–8 MHz. The discs were prepared by pressing the powders in an evacuated die at 12 MPa and then reheating at ~1,050 °C for about 10 h. The discs were then coated with a silver paint and dried at 100 °C, then heated at 600 °C for 30 min to put it between two silver plates to ensure a good electrode/electrolyte contact. The entire cell assembly was placed in a tubular furnace, in which the temperature could be controlled to ± 2 °C with a temperature controller (model YCC-1211) for impedance measurement. Decomposing voltage of the sample was measured by DC polarization experiment using graphite as both of the electrodes in the voltage range of 0–4.0 V in air. All the measurements of electrochemical properties were carried out in ZAHNER Elektrik Thales IM6 Station (Germany).

Results and discussion

X-ray diffraction analysis

Figure 1 shows the powder XRD patterns of the $\text{Li}_{3x}\text{La}_{2/3-x}1/3-2x\text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$) system. No impurity phase is found to exit along with the perovskite phase in the studied composition range of $y \leq 0.10$ except for $y=0$, with TiO_2 exiting as a second phase. Cubic superstructure reflection lines, stemming from the ordering of La^{3+} , Li^+ , and vacancies or alternate arrangement of La-

rich and La-poor layers along C -axis, which nearly doubles c -axis [1, 10–12], are observed in all the studied samples and denoted as asterisks in Fig. 1. From the XRD data, we can find that the strength of the superstructure reflections fluctuate slightly with increasing y value and that not all the superstructure reflection lines change in the same trend. This means that the ordering arrangement of La^{3+} , Li^+ , and vacancies along the c -axis depends on the composition slightly.

With the use of X'Pert Plus to refine the structure of the samples, all the reflection lines can be indexed in tetragonal structure with space group of $P4mm$ (99), as shown in Fig. 1. The calculation results of unit-cell parameters are listed in Table 1.

It is found that the volume of unit-cell decreases with increasing value of y from $y=0.01$ to $y=0.05$, then begins to increase until $y=0.07$. After that, it resumes to decrease to a minimum value. It is very easy to understand that the volume decreases with an increase of y , as the radius of F^- ($r=136$ pm) and Al^{3+} ($r=50$ pm) is smaller than that of O^{2-} ($r=140$ pm) and Ti^{4+} ($r=68$ pm), respectively. With a further increase of the y value, excessive substitution will distort the structure, leading to cell distension in a certain substitution range.

Electrical properties

The ionic conductivity of the $\text{Li}_{3x}\text{La}_{2/3-x}1/3-2x\text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$) system was measured by AC impedance technique. A typical complex impedance diagram is shown in Fig. 2. It is composed of two depressed semicircles in high- to medium-frequency range and a spike in low-frequency range. The spike is the blocking effect of the ionic charge carriers at the electrode–sample interface. The high-frequency semicircle can be ascribed to the bulk behavior, since a capacitance of the order of 100 pF was determined for this semicircle, which is in good agreement with bulk capacitance. The high- to medium-frequency semicircle can be assigned to grain boundary effect, as the capacitance for this semicircle is in the order of nF [5, 14]. It is obvious that grain boundary conductivity is at least two orders magnitude lower than bulk conductivity, which is a

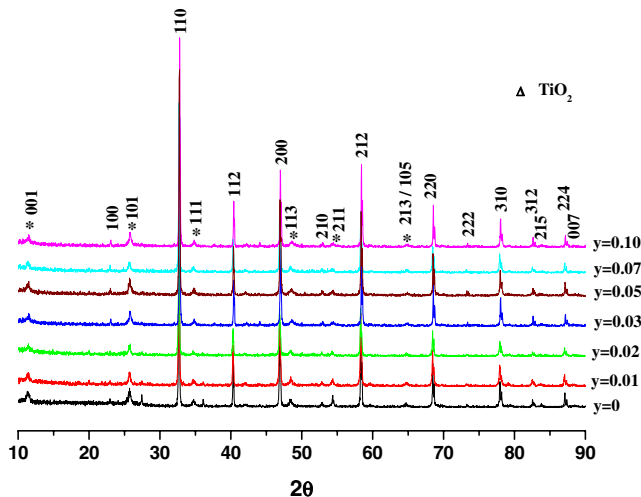


Fig. 1 XRD patterns of $\text{Li}_{3x}\text{La}_{2/3-x}1/3-2x\text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$)

Table 1 Unit-cell parameters of $\text{Li}_{3x}\text{La}_{2/3-x}1/3-2x\text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$) system

Label	y	a (Å)	c (Å)	v (Å ³)
1LTF	0.01	3.8701(7)	7.7560(4)	116.17
2LTF	0.02	3.8697(2)	7.7490(2)	116.04
3LTF	0.03	3.8692(4)	7.7434(7)	115.92
4LTF	0.04	3.8697(4)	7.7440(1)	115.96
5LTF	0.05	3.8667(6)	7.7390(2)	115.71
6LTF	0.06	3.8722(6)	7.7300(5)	115.90
7LTF	0.07	3.8705(1)	7.7476(5)	116.06
8LTF	0.08	3.8695(5)	7.7437(7)	115.95
9LTF	0.09	3.8714(3)	7.7370(1)	115.96
10LTF	0.10	3.8610(1)	7.7360(2)	115.32

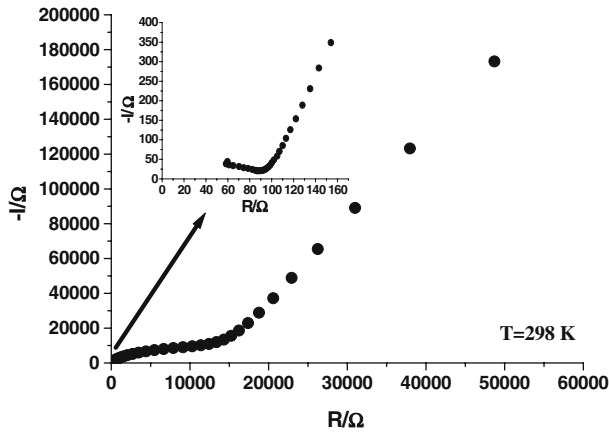


Fig. 2 Complex impedance diagram of 2 LTF with high-frequency part as an *inserted plot*

characteristic of such perovskite ionic conductors that hinders its use for its practical applications [5, 10, 11]. With the use of equivalent circuit to fit the data, bulk conductivity and grain boundary conductivity are determined. The results of bulk conductivity at different temperatures are listed in Table 2.

Figure 3 shows the typical relationship of $\lg(\sigma_b T)$ vs $1,000/T$ for $\text{Li}_{3x}\text{La}_{2/3-x} \text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$). It is found that a classical Arrhenius law is obeyed in this system below 473 K, according to $\sigma T = \sigma_0 \exp(-E_a/RT)$; the activation energy for lithium ion migration are calculated and listed in Table 2. Above 473 K, the Arrhenius plot displays a concave curvature. Such behavior is in accordance with the reports of Belous et al. [15], and it is suggested that such behavior is due to the changing of conductive mechanism from thermally activated to thermally assisted, which may use VTF relationship to describe the relationship between conductivity and temperature [12, 14, 15].

The data in Table 2 show that the sample with initial composition of $y=0.02$ possesses the best ionic conductivity of 1.06×10^{-3} S/cm at RT, which is comparable to the well-known $\text{Li}_{0.34}\text{La}_{0.51}\text{TiO}_{2.94}$ [1], indicating that small substitution of Ti and O by Al and F, respectively, has effectively enhanced the conductivity. The positive effect

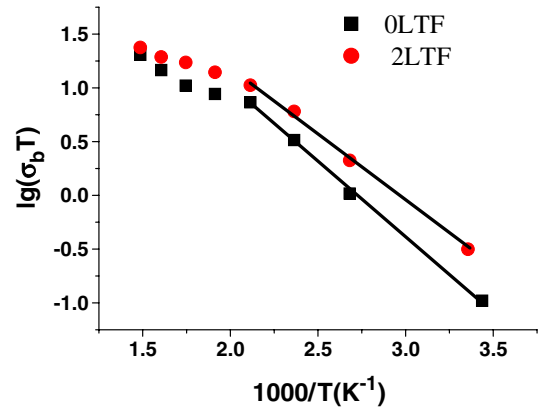


Fig. 3 Plots of $\lg(\sigma_b T)$ vs $1,000/T$ for $\text{Li}_{3x}\text{La}_{0.67-x}\text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$)

of Al substitution has been reported by He and Yoo [10]. To clarify the effect of F substitution, F-doped solid solution $\text{Li}_{3x-0.02}\text{La}_{0.67-x}\text{TiO}_{2.98}\text{F}_{0.02}$ was synthesized by the same method. The results of conductivity measurements show that the bulk ionic conductivity at RT is in the order of 10^{-4} S/cm, with the highest one of 3.17×10^{-4} S/cm for $x=0.11$. These results indicate that doping F solely could not improve the ionic conductivity, but F substitution improves the electrochemical stability of the system. The decomposing voltage of F substitution sample is higher than that of Al,F-codoped sample, as shown in Fig. 4. Thus, Al,F-codoping increases the decomposing voltage while maintaining the high conductivity, and such material shows more potential for application.

To see what will happen when the voltage scans back to 0 V, cyclic voltammetry measurement for sample of 2 LTF between 0–5.0 V was taken. The profile is shown in Fig. 5. When the voltage is lower than 2.0 V, the current is low and exhibits a linear relationship with voltage. Such a low current is mainly contributed by the electronic current and small part of the ionic current. Once the voltage is higher than 2.0 V, the current deviates from the linear relationship, then increases sharply when the voltage is higher than 2.3 V, indicating that the sample started to decompose and lithium ion was reduced to lithium on the interface of the cathode and the sample. Since the measurement was taken

Table 2 Bulk conductivity and activation energy of $\text{Li}_{3x}\text{La}_{2/3-x} \text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$) system

Label	y	σ_b (S/cm)					Ea (kJ/mol), RT-473 K
		RT	373 K	473 K	573 K	673 K	
0LTF	0	3.60×10^{-4}	2.79×10^{-3}	1.55×10^{-2}	1.83×10^{-2}	3.02×10^{-2}	26.7
1LTF	0.01	9.80×10^{-4}	2.10×10^{-3}	2.01×10^{-2}	2.37×10^{-2}	4.26×10^{-2}	24.1
2LTF	0.02	1.06×10^{-3}	5.68×10^{-3}	2.24×10^{-2}	3.01×10^{-2}	3.53×10^{-2}	23.9
3LTF	0.03	9.28×10^{-4}	4.56×10^{-3}	1.73×10^{-2}	2.39×10^{-2}	3.17×10^{-2}	22.7
4LTF	0.04	6.22×10^{-4}	2.68×10^{-3}	9.50×10^{-3}	1.81×10^{-2}	2.37×10^{-2}	21.6
5LTF	0.05	1.76×10^{-4}	1.53×10^{-3}	6.65×10^{-3}	1.05×10^{-2}	1.34×10^{-2}	26.4
6LTF	0.06	1.06×10^{-4}	1.42×10^{-3}	4.09×10^{-3}	5.68×10^{-3}	6.43×10^{-3}	26.5
7LTF	0.07	6.45×10^{-4}	3.19×10^{-3}	1.09×10^{-2}	1.76×10^{-2}	2.28×10^{-2}	22.5
8LTF	0.08	4.94×10^{-4}	1.70×10^{-3}	1.03×10^{-2}	1.67×10^{-2}	2.21×10^{-2}	24.1
10LTF	0.10	1.04×10^{-4}	8.49×10^{-4}	3.23×10^{-3}	5.01×10^{-3}	5.65×10^{-3}	26.6

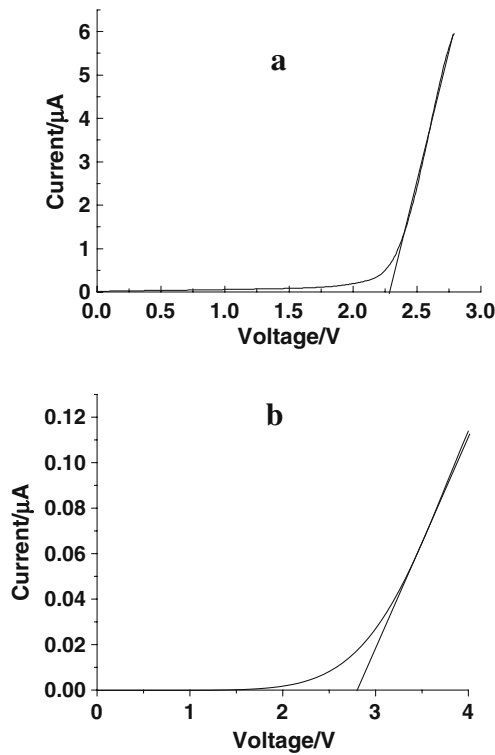


Fig. 4 DC polarizing curve measured in air with sample sandwiched between two graphite electrodes **a** 2LTF and **b** $\text{Li}_{3x-0.02}\text{La}_{0.67-x}\text{TiO}_{2.98}\text{F}_{0.02}$ ($x=0.12$)

in air, lithium then reacted with oxygen and water in the air to produce Li_2O and LiOH , which hindered the ongoing electrolysis of the sample. Thus, the current is kept constant at 3.0–3.3 V and resumes to increase until LiOH begins to decompose. Upon scanning the voltage from 5.00 to 0 V, the ionic current keeps decreasing until only the electronic current remains.

It is found from Table 2 that the ionic conductivity fluctuates with y value, i.e., the ionic conductivity increases when y increases from 0 to 0.02, then it decreases with a further increase of the y value to 0.06. When $y=0.07$, the ionic conductivity reaches another peak value, after which the conductivity resumes to decrease with an increase of the y value. Since the concentration of vacancy and lithium ion does not change with the value of y in the present

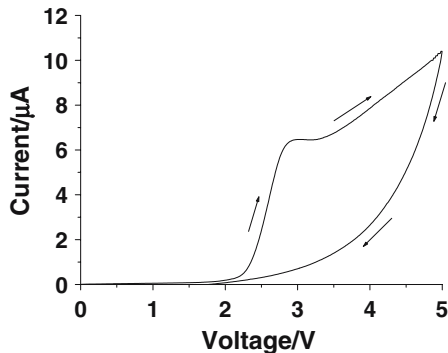


Fig. 5 Cyclic voltammetry profile for sample of 2LTF measured in air with sample sandwiched between two graphite electrodes

system $\text{Li}_{3x}\text{La}_{2/3-x}\text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$), the ratio of vacancy to lithium ion, which may play a role in affecting conductivity, can be ruled out. The ordering of cations and vacancies at A site should not play a major role, as cubic superstructure reflections exit in all the samples and their strengths change slightly with the y value. To explain the changing trend of ionic conductivity, the following factors should be considered:

1. Size of bottleneck. If the value of y does not increase to such a degree that can distort the tetragonal structure, more Ti^{4+} and O^{2-} will be replaced by smaller sized Al^{3+} and F^- respectively, when the y value increases. This leads to the contraction of the unit-cell as we discussed previously, allowing the reduction of bottleneck size for Li^+ migration and making it more difficult for TiO_6 octahedra to tilt [10, 12]. Hence, the ionic conductivity will decrease, but with more Al and F being introduced to the system, the tetragonal structure will be distorted, which makes the volume of unit-cell begin to increase, and a peak value appears at $y=0.07$. Hence, the bottleneck size is enlarged; accordingly, the conductivity reaches another peak value at $y=0.07$.
2. The change of bond strength between A–O and B–O bond in ABO_3 perovskites. It has been suggested that the bond strengths of A–O and B–O in ABO_3 perovskite are competing because they use the same 2p oxygen orbitals [10, 12]. That is, the stronger the A–O bond, the weaker the B–O bond and vice versa, and the strength of an ionic bond is dominated by the interionic distance and the covalent character of the bond [10, 12]. The shorter the interionic distance, the stronger the ionic bond [10, 12]. When smaller Al^{3+} replaces Ti^{4+} , the B–O bond will be shortened. Hence, the B–O bond will be strengthened, weakening the Li–O bond, and the mobility of Li^+ will be enhanced. On the other hand, the polarizing ability of Al^{3+} is slightly weaker than that of Ti^{4+} and Al doping will enhance the ionic character of B–O bond, i.e., enhance the B–O bond and weaken the Li–O bond. As a consequence, the conductivity increases from $y=0$ to $y=0.02$.
3. F is the most electronegative element. The bond between F and other metal elements is mainly of ionic character since the charge of F^- is only half of O^{2-} . The electrostatic force between cations and F^- will be nearly half of that between cations and O^{2-} since the charge of B-site cations is much higher than that of Li^+ . The degree of absolute weakening of B–F bond compared to B–O bond is much larger than that of Li–F bond compared to Li–O bond. Thus, F substitution will enhance the competitiveness of Li–F bond, i.e., strengthen the Li–F bond relatively. Thus, the mobility of Li^+ will be weakened and the ionic conductivity will decrease as a result.

Mainly because of the co-impact of the aforementioned factors, the ionic conductivity shows the fluctuating trend along the y value.

Conclusion

Al,F-doped perovskite lithium ion conductors $\text{Li}_{3x}\text{La}_{2/3-x}1/3-2x\text{Ti}_{1-y}\text{Al}_y\text{O}_{3-y}\text{F}_y$ ($x=0.11$) have been synthesized. Al substitution enhances the ionic conductivity, while F substitution improves the electrochemical stability. The sample with the initial composition of $y=0.02$ possesses the best conductivity of 1.06×10^{-3} S/cm at RT, and its decomposing voltage is 2.3 V. Though from thermodynamic point of view, LLT-based lithium ion conductors cannot be used as electrolyte for batteries using Li as anode [16]. It could be used as electrolyte in lithium ion batteries with open circuit voltage lower than 2.3 V.

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